

Roberto Cantillan Carrasco

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Research Interests

I am a PhD candidate in Sociology at Pontificia Universidad Católica de Chile. My main research examines how the **content of work** has changed over recent decades and how those changes are distributed across the occupational structure. I study how skills, tasks, and other work requirements spread, accumulate, and recede across occupations, jobs, and organizations amid rapid technological change, including advances in artificial intelligence. From a **network science** perspective, I treat these units as interdependent positions: changes in one part of the labor market create opportunities and constraints elsewhere. This approach asks why some kinds of work expand in high-status positions while others become concentrated in lower-status positions, and how these directional shifts contribute to polarization and inequality.

My broader research examines how relational structure shapes social change. Two dissertation projects study how opportunity sets affect the persistence of social ties and when network positions that facilitate one stage of diffusion remain advantageous at another. Emerging work extends this perspective to urban and intergenerational mobility. Across these projects, I combine network analysis, formal models, computational experiments, and longitudinal data.

Education

Pontificia Universidad Católica de Chile

Chile

PhD (candidate) in Sociology

2022 – present

- Dissertation: *Structurally Conditioned Diffusion: How Social and Occupational Structures Filter Propagation*
- Supervisor: Dr. Mauricio Bucca

Universidad de Santiago de Chile

Chile

Master in Social Sciences, mention in civil society studies

2015 – 2018

- Thesis: *Social capital and popular power in territorial mobilization networks*
- Best Thesis Award 2019 — Universidad de Santiago IDEA
- Supervisor: Dr. Vicente Espinoza

Universidad Diego Portales

Chile

Sociology

2007 – 2012

ADVANCED TRAINING:

Complexity Global School for Emerging Political Economies

Colombia

Santa Fe Institute — complex systems, emergence, computational modelling

2025

BEHAVE Summer School on Agent-Based Modelling

Italy

University of Brescia — NetLogo, Mesa, social simulation

2024

SICSS — Summer Institutes in Computational Social Science

Chile

Universidad Católica de Chile — digital trace data, network methods

2023

Research internship

Italy

Università degli Studi di Trento

2019

Summer School in Mixed Methods

Chile

Universidad Católica de Chile

2015

Academic Experience

PEER-REVIEWED PUBLICATIONS:

- Ahumada, G., Cantillan, R., & Jara, B. (2024). Social capital and individual well-being in the post-disaster period: The case of Hurricane Maria in Puerto Rico. *International Journal of Disaster Risk Reduction*, 103, 104308. [doi](#)
- Garreton, M., Espinoza, V., & Cantillan, R. (2021). Social capital in the urban context: Diversity and social contacts in Chilean cities. *Journal of Urban Affairs*, 0(0), 1–20. [doi](#)
- Plaza, A., Cantillan, R., Beck, G., & Espinoza, V. (2022). Together, but not mixed: Homophily in Mapuche and non-Mapuche intergroup relations. *Revista de Sociología*, 37(1). [doi](#)
- Cantillan, R., & Herrera, J. (2020). Actor and block centrality in mobilization organizational networks: The conflict over the Communal Regulatory Plan of Peñalolén, 2011. *Revista EURE*, 46(138). [link](#)
- Plaza Reveco, A., Cantillan Carrasco, R., & Muñoz Rojas, B. (2023). Multilevel analysis for the study of egocentric networks: Applications in the study of homophily in Chile. In *Social Networks: Theory, Methods and Applications in Latin America* (pp. 183–202). CIS. [link](#)

PREPRINTS:

- Cantillan, R., & Bucca, M. (2026). *Structurally Conditioned Diffusion Reproduces Skills-Based Stratification*. arXiv:2602.21369. [preprint](#)

WORKING PAPERS AND MANUSCRIPTS IN PREPARATION:

- Cantillan, R., & Bucca, M. (manuscript in preparation). *Relative Status Directs the Diffusion of Occupational Skill Content*.
- Cantillan, R., & Bucca, M. (manuscript in preparation). *When Do Alternatives Threaten a Tie? Opportunity Sets and Relational Persistence*
- Cantillan, R. (manuscript in preparation). *When Relational Advantage Does Not Carry Across Diffusion Tasks*.
- Bucca, M., Cantillan, R., Molina, M. D., & Pesando, L. M. (manuscript in preparation). *Posh Flight: Emulation from Below and Elite Innovation in a Century of Naming*.
- Ramond, Q., & Cantillan, R. (manuscript in preparation). *Neighborhood peer influence in secondary school choice*.

SELECTED PRESENTATIONS:

- Aug 2026 “A Matthew Effect in Occupational Skill Content: Divergent Diffusion of Cognitive and Physical Skills Reinforces Inequality.” RC28 Summer Meeting, New York University. [slides](#)
- Jan 2026 “Structurally Conditioned Diffusion Reproduces Skills-Based Stratification: How Skill Requirements Propagate — And Why They Preserve Hierarchies.” RC28 International Conference, Sciences Po (online). [slides](#)

- 2025 “Asymmetric Trajectory Channeling in Labor Markets.” [IC2S2](#), Norrköping, Sweden; [INAS](#), Columbia University, New York.
- 2022 SUNBELT Conference, Cairns, Australia.
- 2021 ISA Forum of Sociology; VIII COES International Conference; Cohesive Societies Seminar, Akademie für Soziologie, Germany.
- 2015–2020 RLARS; COES International Conference; DEMOSAL; Chilean Sociology Congress.

RESEARCH PROJECTS (AS COLLABORATOR AND RESEARCH ASSISTANT):

- Pontificia Universidad Católica de Chile** Chile
Research assistant, FONDECYT N° 1261889 — PI: Dr. Mauricio Bucca 2026 – present
 • “New Approaches to the Study of Social Mobility: Expanding Theory, Data, and Methods”
- Universidad Mayor** Chile
Research assistant, FONDECYT N° 11241178 — PI: Dr. Quentin Ramond 2024 – 2026
 • “Segregación residencial y elección escolar en el SAE en Chile”
- New York University Abu Dhabi** Abu Dhabi
Research assistant — PI: Dr. L. M. Pesando & Dr. M. Molina 2023 – present
 • “Trickle-Down Names: How Social Class Shapes Naming Patterns”
- Millennium Nucleus (LM^2C^2)** Chile
Thesis student 2023 – 2025
 • Skill diffusion networks and structural reproduction of labour market stratification — [lm2c2.cl](#)
- Universidad Católica del Maule** Chile
Research assistant, FONDECYT N° 12200560 — PI: Dr. Julien Vanhulst 2022 – 2024
 • “Sustainability Sciences in Latin America: Social network analysis”
- Universidad de Los Lagos — COES** Chile
Research assistant, FONDECYT N° 1190436 — PI: Dr. Óscar Mac-Clure 2021 – 2022
- Centre for Social Conflict and Cohesion Studies (COES)** Chile
Research assistant, MINICOES — PI: Dr. Vicente Espinoza 2021 – 2022
- Universidad de Santiago de Chile** Chile
Research assistant, FONDECYT N° 1171426 — PI: Dr. Vicente Espinoza 2019 – 2022
- Universidad de Chile** Chile
Research assistant, FONDECYT N° 1160984 — PI: Dr. Emmanuelle Barozet 2017 – 2019

STATISTICAL CONSULTANCY:

- FONDECYT 1170324 — PI: Dr. Teresa Correa** 2020
Universidad Diego Portales Chile
 • Panel data analysis
- FONDECYT 1190540 — PI: Dr. Andrés Scherman** 2021
Universidad Diego Portales Chile
 • Experimental data analysis
- FONDECYT 11180916 — PI: Dr. Macarena Peña y Lillo** 2021
Universidad Diego Portales Chile
 • Structural equation modelling and confirmatory factor analysis

TEACHING EXPERIENCE:

Universidad Católica del Maule	Chile
<i>Instructor — Introduction to Computational Social Sciences and Data Science with R</i>	2024
Pontificia Universidad Católica de Chile	Chile
<i>Teaching assistant — Categorical Data Analysis (Master & Doctoral Program)</i>	2023 – 2025
• Director: Dr. Mauricio Bucca	
Pontificia Universidad Católica de Chile	Chile
<i>Teaching assistant — Diploma in Advanced Database Processing in R</i>	2023 – 2024
• Director: Dr. Mauricio Bucca	
Universidad de Santiago de Chile	Chile
<i>Instructor — Thesis Seminar I & II, School of Psychology</i>	2020
Universidad de Santiago de Chile	Chile
<i>Teaching assistant — School of Psychology</i>	2014 – 2019
• Design and evaluation of social projects; Contemporary social psychology; Qualitative methods I and II	
Universidad de Humanismo Cristiano	Chile
<i>Teaching assistant — Thesis Seminar I, School of Psychology</i>	2015

Awards, Grants and Fellowships

NOMAD Data Challenge — Selected Project	2026
<i>Network for Open Mobility Analysis and Data (NOMAD) — University of Pennsylvania</i>	
• Selected proposal: Spatial Functional Bandwidth and Class Mobility Frontiers (with Mauricio Bucca)	
• Competitive selection granting research access to large-scale GPS mobility data and the NOMAD analysis environment	
RC28 Summer Meeting Award	2026
<i>Research Committee 28 (Social Stratification), International Sociological Association</i>	
• Awarded in recognition of scholarship in the field of stratification at the 2026 Summer Meeting, New York University	
Conference Travel Award	2021
<i>International Sociological Association (ISA)</i>	
Master Scholarship	2018 – 2019
<i>Centre for Social Conflict and Cohesion Studies (COES)</i>	
Best Master's Thesis in Social Sciences	2019
<i>Universidad de Santiago IDEA</i>	
Competitive grant for research internship in foreign university	2018
<i>Universidad de Santiago IDEA</i>	

Additional Professional Experience

Universidad Alberto Hurtado	Chile
<i>Data analyst — Institutional Research Unit</i>	2026
• Institutional survey analysis and automated reporting in R and Quarto	
Centro de Estudios Interculturales e Indígenas (CIIR)	Chile
<i>Technical analyst — ELRI longitudinal panel</i>	2023 – 2024
• Survey data management, psychometric assessment, and statistical analysis	

Universidad de las Américas	Remote, part-time
<i>Analyst — Community Intervention Program</i>	2020 – 2022
Centro de Investigación Avanzada en Educación (CIAE)	Chile
<i>Research assistant</i>	2019 – 2020
Universidad de Santiago de Chile	Chile
<i>Analyst</i>	2017 – 2019
Fundación Casa de la Paz	Chile
<i>Assistant</i>	2016 – 2017

Skills and Languages

Statistical computing: R (fixest, lme4, data.table, tidyverse, survey, lavaan), Python, SQL
Reproducibility and reporting: Quarto (automated pipelines), $\LaTeX$, R Markdown, Jupyter Notebooks, Git/GitHub
Infrastructure: Linux/Bash, SSH, DigitalOcean (remote server administration for high-memory computation)
Languages: Spanish (native), English (Professional proficiency)

References

<u>Mauricio Bucca, Ph.D.</u>	Pontificia Universidad Católica de Chile
<i>Assistant Professor, Sociology mebucca@uc.cl</i>	
<u>Mario Molina, Ph.D.</u>	New York University Abu Dhabi
<i>Assistant Professor, Sociology mario.molina@nyu.edu</i>	
<u>Luca Maria Pesando, Ph.D.</u>	New York University Abu Dhabi
<i>Associate Professor, Sociology lucamaria.pesando@nyu.edu</i>	
<u>Vicente Espinoza, Ph.D.</u>	Universidad de la República, Uruguay
<i>profespinoza2010@gmail.com</i>	